

The Most Informative Boolean Function Conjecture Holds for $n = 4$

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Abstract

Let X be uniform on $\{0,1\}^n$ and let Y be the output of a memoryless binary symmetric channel with crossover probability α on input X . Courtade and Kumar conjectured that every Boolean function $f : \{0,1\}^n \rightarrow \{0,1\}$ satisfies $I(f(X);Y) \leq 1 - h(\alpha)$, where h is the binary entropy function: no one-bit quantization of X is more informative about the noisy observation than a single coordinate. We prove the conjecture for $n = 4$ and every $\alpha \in (0, \frac{1}{2})$, with dictators and anti-dictators the unique maximizers on $[0.001, 0.495]$. To our knowledge this is the first rigorous full-noise-range verification for arbitrary Boolean functions at a fixed n . The proof is computer-assisted but self-contained: two short analytic lemmas handle neighborhoods of the degenerate endpoints $\alpha \rightarrow 0$ and $\alpha \rightarrow \frac{1}{2}$, with their finitely many per-class hypotheses verified in exact rational arithmetic, and validated interval arithmetic certifies the strict inequality on the compact middle range, class by class over the 222 NPN equivalence classes of 4-variable Boolean functions. An explicit covering lemma assembles the regimes. All computations are reproducible from the public repository.

Index Terms

Boolean functions, binary symmetric channel, computer-assisted proof, Courtade–Kumar conjecture, interval arithmetic, mutual information.

I. Introduction

How much can a single bit reveal about a noisy vector? Let $X \sim \text{Uniform}(\{0,1\}^n)$ and let Y be obtained from X by passing each coordinate independently through $\text{BSC}(\alpha)$, $\alpha \in (0, \frac{1}{2})$. Every Boolean function $f : \{0,1\}^n \rightarrow \{0,1\}$ is a one-bit quantization of the

Code and verification artifacts: <https://github.com/gokhanmergen/information-theory-problems>.

clean data, and data processing alone bounds its relevance to the noisy observer only by $I(f(X); Y) \leq \min\{1, n(1 - h(\alpha))\}$. Courtade and Kumar [1] conjectured the far sharper

$$I(f(X); Y) \leq 1 - h(\alpha) \quad (1)$$

for every Boolean $f : \{0, 1\}^n \rightarrow \{0, 1\}$, with the dictator functions $f(x) = x_i$ and their complements (anti-dictators) achieving equality: the most informative one-bit summary is a single coordinate. The question arose from CEO-style distributed inference problems [1], and (1) is a fundamental limit on the usefulness of a hard decision, made from clean data, to a remote observer who sees only noise-corrupted data. Beyond its operational content, the conjecture has become a benchmark for the interaction of information theory with the analysis of Boolean functions: a one-line, finite-dimensional statement that has resisted hypercontractivity, Fourier perturbation, and isoperimetric methods for over a decade.

A. Related work

The conjecture remains open in general. Among peer-reviewed results: it holds in the high-noise regime $\alpha \in [\frac{1}{2} - \delta, \frac{1}{2}]$ for an inexplicit absolute constant δ [2]; improved global bounds short of (1) appear in [3]; the conjecture is equivalent to a symmetrized Li-Médard conjecture [6]; and a differential-equation reformulation reduces the balanced case to a finite-dimensional functional inequality [7]. Yu [4] established local optimality of dictators and, combining it with earlier bounds and computer-assisted estimates, confirmed the balanced conjecture for the explicit correlation range $\rho = 1 - 2\alpha \in [0, 0.914]$, uniformly in n . Among preprints, Javanmard and Woodruff [5] sharpened the high-noise entropy expansion to its optimal order and proved a coordinate-wise form of (1) for all Boolean functions, again with inexplicit constants delimiting the proven range. Two claimed full proofs were later withdrawn by their authors [8], [9]. As numerical evidence, Courtade and Kumar [1, Sec. II-B] enumerated compressed Boolean functions through $n = 7$ and evaluated their mutual information; separately, their implementation of Algorithm II.1 sampled a related inequality on a noise grid of spacing 0.001 [1, Sec. II-D]. Neither computation proves (1) on any noise interval.

For arbitrary (including unbalanced) Boolean functions at a fixed n , no proof covering the entire range $(0, \frac{1}{2})$ appears to have been recorded: the high-noise theorems have inexplicit constants, so they cannot be concatenated with grid computations; and the low-noise limit

— where (1) degenerates to equality for every balanced f at $\alpha = 0$ — has no general theorem at all.

B. Contribution

Theorem 1. For $n = 4$, inequality (1) holds for every $\alpha \in (0, \frac{1}{2})$ and every Boolean function f . Moreover, for $\alpha \in [0.001, 0.495]$, equality holds if and only if f is a dictator or an anti-dictator.

The endpoint regimes are exactly where naive certification must fail: the gap $g_f(\alpha) := 1 - h(\alpha) - I(f(X); Y)$ vanishes as $\alpha \rightarrow \frac{1}{2}$ for every f , and as $\alpha \rightarrow 0$ for every balanced f . Our two endpoint lemmas isolate the finite data controlling (1) there.

Endpoint regimes. Edge-boundary profiles and hypercube edge-isoperimetry control the conjecture near $\alpha = 0$; level-one Fourier weight and spectral ℓ_1 -mass control it near $\alpha = \frac{1}{2}$. Each endpoint reduces to finitely many rational inequalities, verified exactly.

Middle range. Validated interval arithmetic certifies strict inequality on the compact middle range, class by class (Section V).

Complete coverage. Lemma 4 assembles the regimes into a gap-free cover of $(0, \frac{1}{2})$. The scheme is dimension-generic; Section VII reports its state at $n = 5$.

Remark 1 (Scope of the uniqueness claim). The interval computation certifies strict inequality for every non-dictator class on $[0.001, 0.495]$, which yields the equality characterization there. The endpoint lemmas establish $g_f \geq 0$ without a uniform separating margin, so uniqueness on $(0, 0.001) \cup (0.495, \frac{1}{2})$ is expected but not certified here; no non-dictator equality case can occur on the certified range, and we know of no mechanism producing one in the endpoint slivers.

II. Setup, symmetry reduction, and class accounting

Throughout, $h(u) = -u \log_2 u - (1 - u) \log_2 (1 - u)$ with the convention $0 \log_2 0 := 0$; $\rho = 1 - 2\alpha$. Write $q_1 = \Pr[f(X) = 1] = |f^{-1}(1)|/16$ and $q_0 = 1 - q_1$; neither depends on α . Since Y is uniform on $\{0, 1\}^4$,

$$g_f(\alpha) = H(f(X), Y)(\alpha) - h(\alpha) - 3 - H(q_1),$$

TABLE I
Disposition of the 222 NPN classes.

Classes	Method	Conclusion
1 constant	analytic: $I = 0$	$g_f = 1 - h(\alpha) > 0$
1 dictator	analytic: $I = 1 - h(\alpha)$	$g_f \equiv 0$ (equality)
220 remaining	Lemmas 2, 3; Proposition 1	$g_f > 0$ (middle), $g_f \geq 0$ (endpoints)

and the 2×16 joint probabilities are the exact polynomials

$$p_{v,y}(\alpha) = 2^{-4} \sum_{d=0}^4 c_{v,y,d} \alpha^d (1-\alpha)^{4-d}, \quad c_{v,y,d} = \#\{x : f(x) = v, d_H(x, y) = d\},$$

where d_H is Hamming distance; this uses that, given $Y = y$, the posterior of X is the product BSC kernel centered at y (Bayes' rule with uniform input and a symmetric channel).

$I(f(X); Y)$ is invariant under permutations of the coordinates, flips of input coordinates, and complementation of the output, so it suffices to prove (1) for one representative of each NPN equivalence class. For $n = 4$ there are 222 classes [10], a count our enumeration reproduces, with orbit sizes summing to 2^{16} . The accounting used throughout is given in Table I. Every computer-assisted step below covers the same 220 classes; the interval certification additionally re-verifies the constant class (whose zero $c_{v,y,d}$ counts drop out of the entropy sum by the $0 \log_2 0$ convention), which is why its logs report 221.

III. The low-noise lemma

For $y \in \{0, 1\}^4$ let b_y be the number of Hamming neighbors x of y with $f(x) \neq f(y)$, and set $c_1 = \frac{1}{16} \sum_y b_y$ (so $16c_1 = 2B$ with B the edge boundary of $f^{-1}(1)$) and $c_2 = \frac{1}{16} \sum_y b_y \log_2 b_y$ (terms with $b_y = 0$ vanish by the $0 \log_2 0$ convention; $b_y \leq 4$ gives $c_2 \leq 2c_1$).

Lemma 1 (code-verified enumeration). Over the balanced NPN classes of Q_4 , the minimum edge boundary of a non-dictator class is 12; dictators attain the edge-isoperimetric minimum 8, uniquely among balanced sets.

The uniqueness of subcubes at the edge-isoperimetric minimum is classical [11], [12], [13], [14]; the value 12 for the best balanced non-dictator is an exhaustive computation over the balanced classes, recorded in the repository (it is not claimed to follow from the isoperimetric theorems).

Lemma 2. Let $\alpha \in (0, 10^{-3}]$, $t = \log_2(1/\alpha)$, $t_0 = \log_2 1000$, $\gamma = (1 - 10^{-3})^3$. If f is balanced and $t_0(\gamma c_1 - 1) \geq \log_2 e + c_2$, or f is unbalanced and $10^{-3}[t_0 \max(0, 1 - \gamma c_1) + \log_2 e + c_2] \leq 1 - H(q_1)$, then $g_f(\alpha) \geq 0$.

Proof. Write $g_f = m_0 - D(\alpha)$ with $m_0 = 1 - H(q_1)$ and $D(\alpha) = h(\alpha) - H(f(X) | Y)$. Three elementary steps:

(i) $H(f(X) | Y=y) = h(\Pr[f(X) \neq f(y) | y])$, and $\Pr[f(X) \neq f(y) | y] \geq b_y \alpha(1 - \alpha)^3 =: b_y \beta$ by keeping only the distance-one terms of the posterior kernel. Both quantities are at most $\Pr[X \neq y | Y=y] = 1 - (1 - \alpha)^4 \leq 4\alpha \leq \frac{1}{2}$, and h is increasing on $[0, \frac{1}{2}]$, so $H(f(X) | Y) \geq \frac{1}{16} \sum_y h(b_y \beta)$.

(ii) $h(u) \geq u \log_2(1/u)$ for $u \in [0, 1]$, hence $h(b_y \beta) \geq b_y \beta [t - \log_2 b_y]$ using $\beta \leq \alpha$.

(iii) $h(\alpha) \leq \alpha(t + \log_2 e)$, from $(1 - \alpha) \ln \frac{1}{1-\alpha} \leq \alpha$.

Combining, and applying $\beta \geq \gamma \alpha$ to the aggregate $c_1 t - c_2$ — which is nonnegative since $c_2 \leq 2c_1$ and $t \geq t_0 > 2$ — while bounding the residual c_2 term with $\beta \leq \alpha$:

$$D(\alpha) \leq \alpha [t(1 - \gamma c_1) + \log_2 e + c_2].$$

If f is balanced then $m_0 = 0$ and $c_1 \geq \frac{3}{2}$ by Lemma 1; the t -coefficient is negative and $t \geq t_0$ gives the stated criterion. If f is unbalanced, αt is increasing on $(0, 1/e)$, so D is maximized at $\alpha = 10^{-3}$, giving the stated criterion against $m_0 > 0$. $\square$

All 220 classes satisfy the applicable criterion, verified in exact rational arithmetic with two-sided rational enclosures of $\log_2 3, \log_2 5, \log_2 7, \log_2 11, \log_2 13, \log_2 e, \log_2 1000$ (each proved by integer exponentiation or a rational series, and used in the unfavorable direction). The minimum slack over all classes is $1680/546377 \approx 3.1 \times 10^{-3}$, attained by class 0x1669 ($q_1 = 7/16$); no class is borderline.

IV. The high-noise lemma

Fourier conventions: for $S \subseteq \{1, \dots, 4\}$ let

$$\chi_S(y) = (-1)^{\sum_{i \in S} y_i}, \quad \widehat{\mathbf{1}}_f(S) = 2^{-4} \sum_x \mathbf{1}_f(x) \chi_S(x),$$

so $\mathbf{1}_f = \sum_S \widehat{\mathbf{1}}_f(S) \chi_S$ and the noise operator satisfies $T_\rho \mathbf{1}_f = \sum_S \widehat{\mathbf{1}}_f(S) \rho^{|S|} \chi_S$ with $T_\rho \mathbf{1}_f(y) = \Pr[f(X) = 1 | Y = y]$. Parseval gives $\sum_S \widehat{\mathbf{1}}_f(S)^2 = 2^{-4} \sum_x \mathbf{1}_f(x)^2 = q_1$ and hence $\sum_{S \neq \emptyset} \widehat{\mathbf{1}}_f(S)^2 = q_1 - q_1^2 = q_0 q_1$. Define $\widetilde{W}_1 = \sum_{|S|=1} \widehat{\mathbf{1}}_f(S)^2 / (q_0 q_1) \in [0, 1]$, $A = \sum_{S \neq \emptyset} |\widehat{\mathbf{1}}_f(S)|$,

and $q_{\min} = \min(q_0, q_1)$. Constants have $q_{\min} = 0$ and are excluded; they were handled in Section II.

Lemma 3. Let $\rho_0 = 10^{-2}$. If $\rho_0 A / q_{\min} \leq \frac{1}{2}$ and

$$[\widetilde{W}_1 + \rho_0^2(1 - \widetilde{W}_1)](1 + \frac{4}{3}\rho_0 A / q_{\min}) \leq 1,$$

then $g_f(\alpha) \geq 0$ for all $\rho \in (0, \rho_0]$.

Proof. Write $p(v, y) = q_v 2^{-4}(1 + \varepsilon_{v,y})$. Since $p(1, y) = 2^{-4}T_\rho \mathbf{1}_f(y)$ and $p(0, y) = 2^{-4}(1 - T_\rho \mathbf{1}_f(y))$,

$$\varepsilon_{1,y} = \frac{T_\rho \mathbf{1}_f(y) - q_1}{q_1}, \quad \varepsilon_{0,y} = -\frac{T_\rho \mathbf{1}_f(y) - q_1}{q_0},$$

and $|T_\rho \mathbf{1}_f(y) - q_1| \leq \sum_{S \neq \emptyset} |\widehat{\mathbf{1}}_f(S)| \rho^{|S|} \leq \rho A$ for $\rho \leq 1$, so $|\varepsilon_{v,y}| \leq \rho A / q_{\min} =: \varepsilon_{\max}$. By Parseval,

$$\chi^2 := \sum_{v,y} q_v 2^{-4} \varepsilon_{v,y}^2 = \frac{\sum_{S \neq \emptyset} \widehat{\mathbf{1}}_f(S)^2 \rho^{2|S|}}{q_0 q_1} \leq \rho^2 [\widetilde{W}_1 + \rho^2(1 - \widetilde{W}_1)],$$

using $\rho^{2|S|} \leq \rho^4$ for $|S| \geq 2$. For $|\varepsilon| \leq \frac{1}{2}$, $(1 + \varepsilon) \ln(1 + \varepsilon) \leq \varepsilon + \frac{\varepsilon^2}{2} + \frac{2}{3}|\varepsilon|^3$: the series $(1 + \varepsilon) \ln(1 + \varepsilon) = \varepsilon + \sum_{k \geq 2} \frac{(-1)^k \varepsilon^k}{k(k-1)}$ is, for $\varepsilon \geq 0$, alternating with decreasing terms after $\varepsilon^2/2$ (negative leading tail term), and for $\varepsilon \in [-\frac{1}{2}, 0)$ all terms are positive with tail at most $\frac{|\varepsilon|^3}{6} \cdot \frac{1}{1-|\varepsilon|} \leq \frac{|\varepsilon|^3}{3}$. Since $\sum_{v,y} q_v 2^{-4} \varepsilon_{v,y} = 0$ (both marginals are exact), summing gives $I(f(X); Y) \ln 2 \leq \frac{\chi^2}{2}(1 + \frac{4}{3}\varepsilon_{\max})$. Finally $1 - h(\alpha) = \frac{1}{\ln 2} \sum_{k \geq 1} \frac{\rho^{2k}}{2k(2k-1)} \geq \frac{\rho^2}{2 \ln 2}$ with all series terms positive. Because $1 - \widetilde{W}_1 \geq 0$, both bracketed factors in the hypothesis are nondecreasing in ρ , so verification at ρ_0 suffices for all $\rho \in (0, \rho_0]$. $\square$

All 220 classes pass, including all balanced ones, with exact dyadic Fourier data; no external high-noise theorem is invoked. The maximum $\widetilde{W}_1$ over non-dictator classes is $52/63$ (a dictator altered on a single input), and the minimum slack of the lemma's hypothesis is 0.1416, attained by that same class; no class is borderline.

V. Validated computation on the middle range

Proposition 1. For every one of the 221 NPN representatives (the 220 classes plus the constant class) and every subinterval recorded in the certification runs below, the outward-rounded interval evaluation of g_f encloses the true range of g_f on that subinterval, its infimum is positive, and the recorded subintervals cover $[10^{-3}, 0.0055] \cup [0.005, 0.495]$ together with the exact-rational bridges $[9/10^4, 11/10^4]$ and $[4949/10^4, 4951/10^4]$.

a) Enclosure validity: The proposition rests on five ingredients:

- NPN reduction correctness: Section II; class count and orbit sum checked against [10].
- Polynomial exactness: the integer-coefficient polynomials $p_{v,y}$ are constructed exactly.
- Enclosure properties: each interval operation used — addition, multiplication, division, and logarithm in `mpmath.iv` at 90-bit precision — is outward-rounded; only these operations are used, and although `mpmath` labels interval support experimental, this elementary-operation path is deterministic and regression-tested in the repository.
- Adaptive bisection: a subinterval is discharged only when the enclosure’s infimum is positive.
- Recorded ledgers for every discharged subinterval.

b) Implementation and runtime: 2029 s and 2.6 s single-threaded (Python 3.14, `mpmath` 1.3.0, macOS/arm64).

c) Reproducibility: The trusted computing base beyond `mpmath` is a ~ 30 -line enclosure routine printed in the repository.

d) Archival status: An immutable archived release with a machine-readable per-subinterval certificate and an independent certificate checker is planned before submission. The audit history, including a repaired floating-point endpoint defect that motivated the two bridges, is in Appendix A.

VI. Covering the noise interval

Lemma 4. Every $\alpha \in (0, \frac{1}{2})$ lies in at least one verified regime, and consecutive regimes overlap on sets of positive measure:

$$(0, \tfrac{1}{2}) = \underbrace{(0, 10^{-3}]}_{\text{Lem. 2}} \cup \underbrace{[\frac{9}{10^4}, \frac{11}{10^4}]}_{\text{bridge}} \cup \underbrace{[10^{-3}, 0.0055] \cup [0.005, 0.495]}_{\text{Prop. 1 runs}} \cup \underbrace{[\frac{4949}{10^4}, \frac{4951}{10^4}]}_{\text{bridge}} \cup \underbrace{[0.495, \tfrac{1}{2})}_{\text{Lem. 3}}.$$

Proof. The union contains $(0, \frac{1}{2})$ by inspection of the rational endpoints; the overlaps $(0, 10^{-3}] \cap [\frac{9}{10^4}, \frac{11}{10^4}]$, $[\frac{9}{10^4}, \frac{11}{10^4}] \cap [10^{-3}, 0.0055]$, $[0.005, 0.0055]$, $[0.4949, 0.495]$, and $[0.495, 0.4951]$ all have positive length. The bridges are certified with exactly represented rational endpoints, eliminating any dependence on the floating-point endpoint handling of the two main runs (Appendix A). $\square$

Theorem 1 follows: constants and dictators are analytic (Section II); every other class satisfies $g_f \geq 0$ on the endpoint regimes (Lemmas 2, 3) and $g_f > 0$ on the middle range

(Proposition 1); and the regimes cover $(0, \frac{1}{2})$ (Lemma 4). The uniqueness statement is Remark 1.

VII. Discussion

The four-regime scheme is dimension-generic. At $n = 5$ there are 616,126 NPN classes — roughly three orders of magnitude more than at $n = 4$ — and the two endpoint lemmas have already been verified for all of them in exact arithmetic (with the low-noise cutover moved to 10^{-4} , as the near-balanced margins shrink); the middle-range certification is embarrassingly parallel by class and is in progress. The genuinely new difficulty as n grows is quantitative: the dictator’s isolation gap shrinks (single flips carry input mass 2^{-n}), so the certified middle range must push closer to the endpoints, where the lemmas’ margins — edge-isoperimetric at low noise, spectral at high noise — grow more robust. Formal verification (e.g., in Lean) of the two finite endpoint checks is a natural hardening step: the per-class conditions are pure rational arithmetic over explicitly listed data.

Contribution and AI-Use Statement

In accordance with IEEE policy on AI-generated content:

Initial generation. The proof strategy, the two endpoint lemmas, all code, and the initial manuscript draft were produced by an AI system (Claude Fable 5, Anthropic) operating on a public open-problems repository.

Independent audit and repair. A separate AI system (GPT-5 Codex, OpenAI) performed adversarial audits that identified the endpoint defect and a citation-provenance correction recorded in Appendix A, and repaired the certification script.

Validation of the proofs. The endpoint lemma conditions and the manuscript’s mathematical claims were independently validated by Gemini 3.5 Flash (Google) and ChatGPT 5.6 Sol (OpenAI).

Human responsibility. The human maintainer supervised the process, reviewed the mathematics, and is responsible for the content. AI-generated material appears throughout the manuscript. AI audit does not substitute for independent human verification; readers are invited to check the two lemmas by hand (they are elementary) and to re-run the public code.

Appendix A

Audit History and Validation

The artifact was adversarially audited before being recorded; two findings are material. First, the script used for the two recorded interval runs converted its decimal endpoint strings to binary floating point before intervalization, rounding 0.001 upward by 2.08×10^{-20} and 0.495 downward by 4.44×10^{-18} ; the recorded runs therefore covered two microscopically shortened intervals. The repaired script keeps all endpoints as exact rationals until evaluation, and the two rational-endpoint bridges of Lemma 4 independently certify neighborhoods of both seams, so the covering argument does not depend on the original endpoint handling. Second, an earlier internal review proposed an alternative high-noise proof by discarding a fourth-order Taylor remainder; that argument has the wrong inequality direction ($h^{(4)} < 0$ makes the corresponding contribution to mutual information positive) and is not used anywhere: the high-noise regime rests solely on Lemma 3. The hard-coded logarithm enclosures used by the exact-rational checks are themselves proved by integer exponentiation (e.g., $2^a \leq n^b$) or a truncated rational series for $\ln 2$, in a self-test that runs before every verification.

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